CURRICULUM VITAE **ROSHNA K**

ADDRESS

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PERSONAL INFORMATION

- **Gender:** Female
- **Date of Birth:** July 04,1996
- **Nationality:** Indian
- **Marital Status:** Married

PRESENT POSITION

- Senior Research Fellow, Department of Physics, National Institute of Technology Karnataka, Surathkal, India, 25.08.2021–Present.

EDUCATION (in reverse chronological order)

Degree	Institution	Year
M. Sc. Physics	Govt College Madappally, Vadakara, India	2019
B. Sc. Physics	Government Brennen College, Thalassery, India	2017

RESEARCH INTERESTS

My research interests include, but is not limited to:

- Primordial scenarios such as inflation, loop quantum cosmology etc.
- Primordial non-Gaussianity and primordial gravitational waves
- Imprints of primordial physics in CMB and Large Scale Structure
- CMB anomalies
- Statistical analysis of models of dark energy

Ph. D. THESIS

Thesis topic: Probing the early and late universe: A phenomenological study with the cosmic microwave background and supernovae

Thesis supervisor: Dr. V. Sreenath

Numerous theoretical insights, together with observational advancements, have enabled us to arrive at the standard model of cosmology, or the ΛCDM model. According to this model, the tiny primordial perturbations generated in the early universe lead to the tiny anisotropies observed in the cosmic microwave

background (CMB). The ΛCDM model, together with inflation, provides a fairly successful explanation for the origin and evolution of these primordial perturbations. Observations of distant type Ia supernovae have shown that our universe is undergoing an accelerated expansion. Within the ΛCDM model, this accelerated expansion can be explained if we assume that our Universe is dominated by dark energy in the form of a cosmological constant. Although the ΛCDM model has been able to explain a wide variety of observations, it raises several questions that challenge our understanding of the Universe. My Ph.D. thesis titled "Probing the early and late universe: A phenomenological study with the cosmic microwave background and type Ia supernovae" investigates two questions concerning the very early and one on the late universe using CMB and type Ia supernovae.

Firstly, Roshna K and V. Sreenath, Phys. Rev. D 113, 023548 (2026) we study two anomalies observed in the CMB, namely suppression of power at large angular scales and the lensing anomaly. Even though individually these anomalies show only a mild deviation from the predictions of the ΛCDM model, their combined occurrence may be significant enough to indicate physics beyond the ΛCDM model. We investigate the existence of power suppression and subsequent alleviation of lensing anomaly, using some simple phenomenological extensions to a nearly scale-invariant primordial power spectrum. We perform Bayesian analysis of these models with the Planck and non-CMB datasets and show that models favouring stronger power suppression also lead to a reduction of the lensing anomaly. We further show that the future near-ultimate CMB missions, such as ECHO, which aim to measure polarization to great accuracy, can put tighter constraints on these models and thereby ascertain the presence of power suppression at low multipoles.

Even though inflation is successful, it is past incomplete as it does not address the initial big bang singularity. Loop quantum cosmology (LQC) is an effort to incorporate the planck scale physics into the standard model of cosmology. In LQC, quantum gravitational effects in the planck regime replace the big bang singularity with a quantum bounce. Studies show that the perturbations generated in LQC are highly non-Gaussian in nature. The dimensionless function $f_{\rm NL}(k_1,\, k_2,\, k_3)$ which quantifies the amplitude of non-Gaussianity generated in LQC, though similar to that generated during slow-roll inflation at small wavelengths, is highly scale-dependent and oscillatory at long wavelengths. The second analysis Roshna K and V. Sreenath, JCAP 08, 014 (2023) we carried out is to compute the imprints of the highly oscillatory, scale-dependent bispectrum generated in LQC on the CMB reduced bispectrum. To perform this calculation, we propose an analytical template for the primordial bispectrum in LQC. This template consists of a strongly scale-dependent, oscillatory part that captures the effect of bounce, and a scale-invariant part that captures behaviour similar to that of slow-roll. We compute the corresponding reduced bispectra of temperature and electric polarization, as well as their three-point cross-correlations. We observe that the contribution from the bounce to the CMB reduced bispectrum is negligible compared to the scale-invariant part. Hence, the reduced bispectrum generated in LQC will be similar to that in slow-roll inflation and thus consistent with observations by Planck.

Finally, we constrain the nature of dark energy by employing frequentist resampling techniques to the latest PantheonPlus and SH0ES (PPS) supernovae datasets Roshna K et. al., arXiv:2601.16197v2 [astro-ph.CO]. While most of the supernovae analyses have relied on Bayesian techniques, frequentist methods have received comparatively less attention. In this work, we employ resampling techniques, namely Jackknife and Bootstrap, together with generalised least squares, to constrain dark energy models such as flat ΛCDM, ΛCDM, flat wCDM, wCDM, and flat $w_0\, w_a$CDM models using the PantheonPlus and SH0ES datasets and compare them with Bayesian methods, viz. Markov chain Monte Carlo and nested sampling. To assess the robustness of these techniques, we calibrate them by applying them to a near-ideal mock fiducial ΛCDM and other models with a one-percent Gaussian error, and then to three versions of the PantheonPlus data. We find that the Hubble tension appears less significant with the Jackknife method. We observe that constraints obtained with different techniques for three- and four-parameter models are largely consistent. Our results highlight the importance of using multiple techniques while analyzing data and warrant further investigation.

PUBLICATIONS AND PREPRINTS

1. **Roshna K**, Nikhil Fernandes, P Praveen, V. Sreenath, *Constraining dark energy models using Jack-*

knife and Bootstrap resampling, arXiv:2601.16197 [astro-ph.CO],

2. **Roshna K** and V. Sreenath, *Power Suppression and Lensing Anomaly – A phenomenological investigation*, arXiv:2504.21386 [astro-ph.CO], Phys.Rev.D 113 (2026) 2, 023548.

3. **Roshna K** and V. Sreenath, *Viability of loop quantum cosmology at the level of bispectrum*, accepted in the Proceedings of the Seventeenth Marcel Grossmann Meeting held at 'Gabriele d'Annunzio' University, Pescara, Italy from 7 - 12 July, 2024, arXiv:2410.00614 [astro-ph.CO].

4. **Roshna K** and V. Sreenath, *Estimation of imprints of the bounce in loop quantum cosmology on the bispectra of cosmic microwave background*, JCAP 08 (2023) 014 , [arXiv: 2301.05406 [gr-qc]].

COMPUTATIONAL SKILLS

- I have worked with Boltzmann codes such as `CLASS` and `CAMB`.
- I have experiance in handling software packages for performing Bayesian analysis in cosmology such as `CosmoMC`, `Cobaya`, `MontePython` and `PolyChord`.
- I have experience working with recent cosmological datasets, including the `Planck PR4` release and the `PantheonPlusSH0ES` compilation of Type Ia supernovae, and incorporating them into cosmological likelihood analyses.
- I have experience with `GetDist` for MCMC chain analysis and posterior visualisation.
- I am familiar with `healpy` Python package for CMB map-making and pixelised data handling.
- I have the ability to build and compile, packages and external libraries.
- I am familiar with the programing languages: Python and C.
- I am comfortable in working in linux based operating systems and high-performance computing (HPC) environments.

PARTICIPATION IN WORKSHOPS AND CONFERENCES (in reverse chronological order)

1. *Cosmic Connections- Second Edition: AI/ML Across Cosmic Scales*, *Institute of Mathematical Sciences (IMSc), Chennai, September 7 - 11, 2026.*
2. *Cosmic Origins and the Search for New Physics* , *Indian Institute of Technology (IIT) Madras, Chennai, India, January 19–23, 2026.*
3. *Cosmic Connections: Bridging the Early and Late Universe* , *Institute of Mathematical Sciences (IMSc), Chennai, July 28 - August 1, 2025*
4. *17th Marcel Grossmann Meeting* , *Aurum, the 'Gabriele d'Annunzio' University and ICRANet, Pescara, Italy, July 07-12, 2024 (Attended Online).*
5. *CosmoGravitas: 1st International Conference on Cosmology and Gravitation* , *Centre for Theoretical Physics and Natural Philosophy Nakhonsawan Studiorium for Advanced Studies, Mahidol University, Nakhonsawan Campus, Thailand, June 10-14, 2024.*
6. *9th Southern Regional Astronomy Meeting - "Research in Astronomy: Opportunities and Challenges" (RAM 2024), Manipal Centre for Natural Sciences (MCNS), Eshwar Nagar, Manipal, Karnataka, India, January 10-12, 2024.*
7. *10th International Conference on Gravitation and Cosmology: New Horizons and Singularities in Gravity, IIT Guwahati, India, December 06 - 09, 2023.*

8. *8th Southern Regional Astronomy Meeting - "Research in Astronomy: Opportunities and Challenges", Department of Basic Sciences and Humanities, Rajagiri School of Engineering and Technology, Kakkanad, Kerala, Febrauary 17 - 19, 2023.*

9. *32nd meeting of Indian Association for General Relativity and Gravitation (IAGRG32), Indian Institute of Science Education and Research Kolkata, India, December 19 - 21, 2022.*

10. *Physics of the Early Universe, attended online at the International Center for Theoretical Sciences, Bengaluru, India, January 3 - 12, 2022.*

11. *LISC Continuous Gravitational Waves Workshop , held online at International Centre for Theoretical Sciences, Bengaluru, India, October 25 - 27, 2021.*

12. *Physics Training and Talent Search (PTTS-2018), at National Institute of Technology, Surathkal.*

TALKS (in reverse chronological order)

1. *"Constraining flat ΛCDM model using Jackknife and Bootstrap resampling"* in Cosmic Connections-Second Edition: AI/ML Across Cosmic Scales , Institute of Mathematical Sciences (IMSc), Chennai, September 7 - 11, 2026.

2. *"Power suppression and lensing anomaly - A phenomenological investigation and predictions for ECHO"* in Cosmic Origins and the Search for New Physics , Indian Institute of Technology (IIT) Madras, Chennai, India, January 19–23, 2026.

3. *"Power Suppression and Lensing Anomaly – A phenomenological investigation"* in Cosmic Connections: Bridging the Early and Late Universe, Institute of Mathematical Sciences (IMSc), Chennai, July 28 - August 1, 2025

4. *"Power Suppression and Lensing Anomaly – A phenomenological investigation"*, at Raman Research Institute (RRI), C. V. Raman Avenue, 5th Cross Road, Sadashivanagar, Near Mekhri Circle, Bengaluru, July 17, 2025.

5. *"Viability of loop quantum cosmology at the level of bispectrum"* in 17th Marcel Grossmann Meeting , Aurum, the 'Gabriele d'Annunzio' University and ICRANet, Pescara, Italy, July 07-12, 2024 (Attended Online).

6. *"LQC: non-Gaussian yet Gaussian"* in 1st International Conference on Cosmology and Gravitation , Centre for Theoretical Physics and Natural Philosophy Nakhonsawan Studiorium for Advanced Studies, Mahidol University, Nakhonsawan Campus, Thailand, June 10-14, 2024.

7. *"Consistency of loop quantum cosmology with the cosmic microwave background bispectrum."* in 9th Southern Regional Astronomy Meeting - "Research in Astronomy: Opportunities and Challenges" (RAM 2024), Manipal Centre for Natural Sciences (MCNS), January 10-12, 2024.

8. *"Consistency of loop quantum cosmology with the cosmic microwave background bispectrum."* in 10th International Conference on Gravitation and Cosmology: New Horizons and Singularities in Gravity, IIT Guwahati, India, December 06 - 09, 2023.

9. *"Imprints of primordial non-Gaussianity in the Cosmic Microwave Background from Loop Quantum Cosmology"* in 8th Southern Regional Astronomy Meeting - "Research in Astronomy: Opportunities and Challenges", Department of Basic Sciences and Humanities, Rajagiri School of Engineering and Technology, Kakkanad, Kerala, February 17 - 19, 2023.

10. *"Imprints of primordial non-Gaussianity in the Cosmic Microwave Background from Loop Quantum Cosmology"* in 32nd meeting of Indian Association for General Relativity and Gravitation (IAGRG32), Indian Institute of Science Education and Research Kolkata, India, December 19 - 21, 2022.

POSTER PRESENTATION

1. *"Consistency of loop quantum cosmology with the cosmic microwave background bispectrum."* in Young Astronomers Meet 2024 funded by Astronomical Society of India and organized by Department of Physics and Electronics, CHRIST(Deemed to be University), Banglore, 06-09 March 2024.

GRANTS AND FELLOWSHIPS

- National Institute of Technology Karnataka, Surathkal, institute Ph.D. fellowship, 2021-2026.
- I received the DST INSPIRE Scholarship (2014-2019) during my undergraduate and master's studies.

SYNERGISTIC ACTIVITIES

- Organized and coordinated biweekly seminars for two years under the "Physics forum" of the Department of Physics, NITK.

TEACHING

At National Institute of Technology Karnataka, Surathkal, I have taught the following courses:

1. Physics Lab (for first year engineering undergraduate students)
2. General Physics Lab (for M. Sc. students)
3. Tutorial Class (for first year engineering undergraduate students)

DETAILS OF REFERENCES

1. Dr. V. Sreenath
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   E-mail: sreenath@nitk.edu.in
2. Dr. Ajith K M
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   E-mail: ajith@nitk.edu.in
3. Dr. T. K. Shajahan
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